Copyright © 1998. Taylor & Francis Group. All rights reserved.

Chapter 6

On interactivity

Consumers, citizens and culture

Andrew Barry

The relation between the contemporary science museum and its public is an interactive one. At the level of the institution, the museum is increasingly expected to respond to the publics demands rather than simply tell the public what it needs to know—the public needs to understand science but, before this is possible, the museum must first understand what the public wants. At the level of the gallery, museum staff aim to design exhibits which enable visitors to make choices and to experience a gallery 'in their own way' (Macdonald and Silverstone 1990:184).[1] And at the level of the individual display, the museum seeks to develop and employ techniques which encourage greater dialogue with the visitor. As an influential Management Plan for London's Science Museum noted:

Passive and poorly interpreted attractions will suffer at the expense of those that develop live demonstrations, provide participation, interactive displays, and give a quality of personal rather than institutional service to their visitors. Informality and friendliness will be valuable attractions.

(Science Museum 1986)

In exploring the proliferation of forms and techniques of interactivity in the museum of science this chapter develops three themes. The first concerns the contemporary political resonances of interaction. There is no doubt, as Mark Poster notes, that the usage of the idea of interactivity can 'float and be applied in countless contexts having little to do with telecommunications' (Poster 1995: 33). Yet in the museum of science, interactivity can have a particular significance, drawing together concerns with, for example, both public 'participation', 'empowerment' and 'accountability' and with more specific questions and anxieties about the proper way to bridge the gulf between popular culture and the esoteric world of science and technology.

The contemporary concern with interactivity emerges at a moment when there is perceived to be a crisis in the relations between science and the public. The public are thought to be ambivalent about the

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

authority of science and, at the same time, are said to be uninformed; a perception which ironically may serve, as Brian Wynne observes, to encourage even ‘more public alienation, hence justifying and consolidating the neuroses’ (Wynne 1992:281; see also Wynne 1993). Some argue that interactive technologies can contribute to the solution of the problem of public misinformation by giving the public a practical and a creative understanding of what it is to be a scientist. Promoting a belief in the value of science to the wider social order depends, in this view, not just on the ritual public display of technological achievement, but on the development of the creative scientific abilities of the individual citizen.

The second theme of this chapter concerns the issue of interactivity as a theoretical concept. The idea of interactivity is one of a number of terms (including noise, feedback and network) which have acquired particular significance since the development of communications theory and cybernetics in the 1940s with the work of, amongst others, Norbert Wiener. In the cybernetic account there is no essential distinction between the capacities of the human and the non-human actor. Both the human and the machine act as sources and receivers of information thereby functioning as part of an interacting system. As Peter Galison reminds us, ‘according to the cyberneticist, the world is nothing more than the internal relations of these incoming and outgoing messages’ (Galison 1994:255–6). Within the contemporary science museum, the technology of interactivity can be intended, if not necessarily to obliterate, at least to reconfigure the distinction between the human visitor and the non-human exhibit.

My final concern is with what we might call, following Foucault, the *political anatomy* of the museum visitor (Foucault 1979:138; see also Schaffer 1992: 329). Although the interest in interactivity has to be seen in relation to the development of communications theory and technology, it can also be located in relation to a rather more longstanding concern with the body as source of experimental knowledge. Simon Schaffer notes that in the eighteenth and early nineteenth centuries the body of the natural philosopher, or of his audience, frequently functioned as an essential part of the experimental apparatus. The eighteenth-century Parisian lecturer Jean Antoine Nollet, for example, ‘described “beatifying electricity”, when sparks were drawn from victims’ hair’ (Schaffer 1992:333). In 1800 the English chemist Humphry Davy reported on his experiments with the inhalation of gas. Davy ‘lost all connection with external things… I existed in a world of newly connected and newly modified ideas. I theorised—I imagined that I made discoveries’ (ibid.: 359).

Since the late nineteenth century, the significance of a scientists body to experiment has changed. The body of the practising scientist has become disciplined, capable of performing meticulous practical tasks

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

and making exact observations but no longer serving as an experimental instrument in itself. The process of science education is, at least in part, a matter of turning the untutored body of the student into that of a reliable technician. As John Law observes, the discipline of the scientist's body can play an important role in laboratory work (Law 1986:21). Experimental events are no longer *experienced* by the scientist; they are *recorded by* the scientists' instruments (Barry 1995). By contrast, the relatively undisciplined body of the visitor has an increasingly important part to play both in the contemporary science museum and what is often called 'the science centre'.[2] Today, the visitor to the museum or the science centre is often encouraged to interact or to 'play' with an exhibit. In effect, the visitor is expected to make scientific principles visible to themselves through the use of touch, smell, hearing or the sense of physical effects on their own bodies (Durant 1992:8). In a manner foreign to the practice of contemporary experimental science, the body is itself a source of knowledge (Gregory 1989:4).

In exploring the significance and recent history of interactivity, this chapter focuses on two contemporary science museums or science centres; the Science Museum in London and the Cité des Sciences et de l'Industrie at La Villette in Paris.[3] In many ways these institutions are unique. In London, interactivity has developed in a museum which has evolved gradually over a long period of time and carries with it a complex historical legacy. This legacy consists not only of rich collection of material artefacts, but also a history of ways of displaying and visiting which goes back to the nineteenth century. In the first part of the chapter I focus on the key role that interactive devices are expected to play in breaking the museum away from what are regarded as some of the more problematic features of this past.

By contrast to the Science Museum, the Cité is a recent development, and the technology of interactivity is integral to the institutions conception and design. Whereas the Science Museum houses a vast collection of historical artefacts, the Cité displays little of the traditional curatorial interest in collection and the history of technology. In the second part of this chapter, I argue that interactive devices function in the Cité, not just as a presentational devices, but as a key element of the institutions vision of the technological present.

CONSUMERS AND CITIZENSHIP: THE SCIENCE MUSEUM AND THE EXPLORATORIUM

In Britain, the new emphasis on 'interactivity' can be understood, in part, in relation to broader changes in the public function of the museum. The modern science museum originally developed in the nineteenth century as a place where the successes of the imperial state could be displayed and where

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

'European productive prowess was typically explained as a justification for empire' (Bennett *et al* 1993: 59).[4] But it was also intended to be a liberal space within which a bourgeois public would participate, and be seen to participate, in their own cultural and moral improvement. Thus, the population would be managed, as Tony Bennett argues, 'by providing it with the resources and contexts in which it might become self-educating and self-regulating' (Bennett 1995:40). As an institution of government, the museum would act not so much through controlling and disciplining the public, but by enlisting its active support for liberal values and objectives.

Museums and expositions, in drawing on…techniques and rhetorics of display and pedagogic relations…provided a context in which working and middle-class publics could be brought together and the former—having been tutored into forms of behaviour to suit them for the occasion—could be exposed to the improving influence of the latter.

(ibid.: 86)

The complex of museums developed at South Kensington in the 1850s became the paradigm of this liberal exhibitionary strategy (ibid.: 40).

In recent years, however, the liberal conception of culture as a means of individual improvement has had to run alongside—if not compete with—neo-liberal notions of culture as a consumer product (Silverstone 1992:41). The traditional museum has been accused of being too paternalist, too dominated by the concerns of curators and the fetishism of the artefact, and too dependent upon public subsidy.[5] What is said to be required is a new recognition of the competitive character of the visitor business *in addition to* the older preoccupations with scholarship and public education. The museum is but a 'part of the leisure and tourist industries' (Kirby 1988:91).

For Neil Cossons, director of the London Science Museum in the late 1980s and 1990s the new concern with the customer was a challenge that should be welcomed, for, with the decline of state funding, 'spending power, and therefore choice, [would be put] into the hands of the people'. For the museums, according to Cossons, the implications were clear: 'The battlefield will be the marketplace and the casualties will be those museums that fail to adapt' (Cossons 1987:18). Seen in this context, 'interactivity' has a double function. First, it is one of a range of technical methods—along with cost

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

control, visitor research, quality assurance, marketing and customer relations—which would enable the museum to forge a more 'economic' relation both with its visitors, and with private industry.[6] As one commentator has noted:

for interactive media the combination of: 1. multiplexing as a delivery mode; 2. interactivity as an intrinsically engaging form of media; 3. niche marketing as an advertising strategy; 4. the affluent status of museum visitors as a demographic group; 5. museums' status as pillars of respectability on scientific, environmental and heritage issues…will greatly increase the attractiveness of museums to sponsors.

(Nash 1992:184)

Second, the technology of interactivity had a function in the context of broader changes in political thinking on both the Left and the Right. Contemporary political thinking is increasingly sceptical of the political and economic competence of the State and, in its stead, relies on the self-governing capacities of the individual, the family, the enterprise or the community (Barry, Osborne and Rose 1996). As Nikolas Rose observes, the subject of what he calls 'advanced' liberal forms of government is given unprecedented responsibility for governing his or her own affairs. For advanced liberalism, the task of the public authorities is not to direct or provide for the citizen but to establish the conditions within which the citizen could become an 'active agent in his or her own government' (Rose 1993:296). Seen in this context, interactive devices have a function, for they may foster agency, experimentation and enterprise, thus enhancing the self-governing capacities of the citizen. Interactivity promises, in other words, to turn the museum visitor into a more active self (cf. Strathern 1992:41–3, Macdonald 1993).

The association of the idea and techniques of interactivity with a broader conception of the public function of the science museum was not new. In 1969, the nuclear physicist Frank Oppenheimer, who had been blacklisted from practising as a scientist by the House Un-American Activities Committee, established the Exploratorium in San Francisco as an alternative to the traditional science museum. For Oppenheimer, existing museums in the United States often glorified the achievements of earlier scientists at the expense of enabling visitors to engage in a process of discovery themselves.[7] The radical message of the Exploratorium was one of *democratic empowerment* (cf. Cruikshank 1996). The public would be empowered through being able to *interact* with objects as an experimental scientist does in the natural world of the laboratory. According to Hilde Hein, 'interactive pedagogic technique contains a key to empowerment that could transform education on a broad scale and make an avenue of general self-determination' (Hein 1990: xvi). In short, visitors would be participants rather than mere observers.

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

Increasingly concerned about the growing interest in mysticism, drugs and Eastern religions in the younger generation, Oppenheimer himself expressed the intellectual and political aspirations of the Exploratorium in these terms:

The whole point of the Exploratorium is to make it possible for people to believe that they can understand the world around them. I think a lot of people have given up trying to comprehend things, and *when they give up with the physical world they give up with the social and political world as well*

(ibid.: xv, emphasis added)

If the idea of interaction was central to what the Exploratorium was trying to do, how was it possible to realize this in practice? What was an interactive technique and how could interactivity *empower?* In the early years, the Exploratorium's attempts to develop participatory and interactive exhibits were, no doubt, rudimentary. The Exploratorium staff had, themselves, to learn how to embody Oppenheimer's radical philosophy in a technical form. However, the centre was able draw on and translate other models of interactivity. One was a temporary exhibition of *Cybernetic Serendipity* which had originally been shown at the Institute of Contemporary Arts in London from August to October 1968 and which was designed to explore the relations between creativity and 'cybernetic' technologies such as computers, robots and mechanical feedback systems. For Oppenheimer and the Exploratorium the origins of the *Cybernetic Serendipity* as an art exhibition accorded with the centre's modernist philosophy. First, neither Oppenheimer nor the exhibition organizers perceived a fundamental distinction between art and science. For Oppenheimer, science had an 'aesthetic dimension and art and science were united in the 'human quest for understanding' (ibid.: xvi). According to the exhibition organizers, 'at no point was it clear to any of the visitors walking around the exhibition, which of the various drawings, objects and machines were made by artists and which made by engineers; or, whether the photographic blow-ups of texts mounted on the walls were the works of poets or scientists' (Reichardt 1971:11). Blurring the boundaries of art and science was an important part of the Exploratorium's pedagogic strategy, for by doing so it was hoped that the centre's visitors might begin to understand that science was a *creative* activity.

In terms of the Exploratorium's philosophy, a second positive feature of the ICA exhibition was that it engaged with science at the level of material practice rather than merely at the level of metaphorical association. Visitors could have a practical as well as visual experience of technology. The instructions to one of the exhibits of *Cybernetic Serendipity*, for example, invited the museum visitor to interact with

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

a machine by turning knobs that adjust the phase and frequency of two wave oscillations relative to one another in order to produce a variety of patterns (Hein 1990:38). Thus, the relationship between scientific or mathematical truth and art would, through a process of interaction, be revealed to the uninitiated. This philosophy still persists. Addressing what he perceives to be the 'crisis in science education, the new director of the Exploratorium, Dr Goéry Delacôte has created a 'Centre for Public Exhibition' which 'provides informal science education through interactive exhibits which address and explore the relationship between science, art and human perception' (Délacôte 1992).

A further intellectual rationale for the idea of the interactive exhibit in the Exploratorium was found in the work of the psychologist Richard Gregory. In his Royal Institution Christmas lectures of 1967/8, Gregory had expounded a theory which held that visual perception entailed a complex integration of the perceivers' interpretive dispositions with external stimuli. According to Gregory,

perception is not a matter of sensory information giving perception and guiding behaviour directly, but rather the perceptual system is a 'look up' system; in which sensory information is used to build gradually, and to select from, an internal repertoire of 'perceptual hypotheses'.

(Gregory 1970:174)

Translating this into practical terms, the Exploratorium 'let the visitor be the laboratory subjects of their own perceptual experiments' (Hein 1990:72). The intended effect of this pedagogic strategy was not just to teach perceptual theory, but to encourage the visitor to experience the process of discovery and thus to become an experimenter.

The extraordinary enthusiasm for interactivity, which had been initiated by the Exploratorium in the late 1960s and subsequently spread across the United States, finally arrived in Europe in the mid-1980s with the opening of the Launch Pad gallery at the London Science Museum, the Cité des Sciences et de l' Industrie in Paris and Richard Gregory's own Exploratory in Bristol. By the early 1990s the growth of interactive science exhibits in Britain had been phenomenal (Stevenson 1994:30). Indeed, a veritable interactives movement had emerged with the formation of associations such as the British Interactives Group (BIG) and the European Collaborative Science, Industry and Technology Exhibitors (ECSITE). Curators, educationalists and museum managers began to share their ideas about the function and design of interactives and encourage the use of interactives in exhibition spaces in which they had previously not been found, such as art galleries.

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

If the Exploratorium provided a model for the interactives movement in Britain it would be a mistake to imagine that the interactivity in Britain was simply a copy of the American original. As sociologists of technology have been at pains to argue, the process whereby a technology is 'transferred' from one place to another should be thought of as a form of translation or reinterpretation rather than merely a form of diffusion (Latour 1986). In the UK, the radical concerns of the American centre with the issue of empowerment were marginalized and, with exceptions, Oppenheimer's interests in the links between science and art were ignored. Instead, interactivity came to operate in relation to the failure of the traditional science museum to address a rather more mundane set of concerns with the public understanding of science and the attractiveness of the museum to visitors.

Criticism of the traditional science museum was most forcefully made by Richard Gregory, founder of the Bristol Exploratory and former adviser to the Exploratorium. For Gregory, 'looking at the traditional museums of science we find remarkably little science' (Gregory 1989:7). Gregory saw the the essential feature of science as experimentation, so that in order to enable the public to get an 'intuitive feel for…the principles of science' hands-on interactive experiences were, he believed, critical. 'I suggest', wrote Gregory, 'that the major aim of interactive science centres, after stimulating interest and curiosity, should be setting up hand-waving explanations giving useful intuitive accounts' (ibid.: 5).

The new interactive science centres were certainly popular. 'Science centres attract visitors like magnets', noted the education officer at the Science Museum responsible for Launch Pad (Stevenson n. d.: 18). Interactive exhibits, whether located within science centres or in more traditional object-centred exhibitionary spaces were also consistently rated highly by the public according to visitor research. According to the Science Museums own research Launch Pad receives 714 visitors per square metre of gallery space per year, while the entire museum receives only 44 visitors per square metre per year (Thomas n.d.: 3).[8] This popularity has proved both a benefit and problem for the development of interactive exhibitions. Certainly, the fact that interactives are popular is of considerable commercial and political significance in a period in which the museum or the science centre is increasingly understood as one part of a broader leisure industry and where the traditional curatorial concern with collection has been downplayed. Accusations that interactive science centres are merely expensive playgrounds that convey little of the tedious and difficult reality of science can be met with the response that this is what the public wants.[9] If interactive galleries enable the public to have fun and to enjoy some kind of experience of science then, in this view, that is sufficient justification for their development. In a period

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

in which visitor numbers are taken to be one of the key performance indicators used in museums, then the case for increasing the space given to interactive exhibits within the museum can appear unanswerable (cf. Thomas n.d.).

However, the recognition that visitors came to interactive science centres and exhibitions to enjoy themselves created a problem for proponents of interactivity (Gregory 1989:2). In the view of their designers, interactive exhibits were always expected to be as much instruments of informal education as a means of entertainment. The museum visitor was conceived of as an active learner, and not just as a consumer. Critics pointed to the lack of historical or industrial contextualization of many interactive exhibits and the frequent absence of any explanation of what scientific principles were supposed to be revealed through the process of interaction. Some exhibits, it was said, can be interpreted in ways which lead museum visitors to false conclusions.[10] Indeed, it is unclear whether any of the scientific principles that many interactives are meant to demonstrate would be grasped by any except those already possessing a good scientific education.[11] Moreover, some question whether many interactive devices are really interactive. Many so-called interactive touch-screen computers, for example, simply allow the visitor to select from a predetermined set of options. Far from providing the possibilities for experimentation, such interactive devices may merely serve to create the illusion of choice (Strathern 1992:42).

There are, no doubt, many different interpretations and responses to this complex conjuncture. At the London Science Museum large numbers of human ‘explainers’ are employed to make sure that the interactive exhibits do the job they are intended to do. New galleries such as Health Matters and the temporary exhibition space, Science Box, incorporate increasingly sophisticated interactive exhibits as a matter of course.[12] There is now more emphasis on tailoring interactives to particular age-ranges so that it is possible to act on the specific technical competencies and interests of the young visitor. In these circumstances, gallery designers increasingly draw on the diverse insights of visitor research, ergonomics, sociology, developmental psychology and educational theory to ensure that the new interactive galleries prove to be educational as well as entertaining.[13] To an extent unparalleled in the past, the museum visitor has become the object of investigation. If interactive technologies are expected to enhance the agency of the visitor and to channel it in the most productive direction, then the specific dynamic of this agency must itself be known (cf. Foucault 1979). The visitor, it seems, has been increasingly called on to interact with exhibits and respond to the growing number of explainers, actors and researchers who also inhabit the museum.

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

CYBORGS AND CULTURE: THE CITÉ

In what follows I want to look at a museum where ‘interactivity’ has a rather different set of resonances. At the Cité des Sciences et de l’Industrie at La Villette in Paris, ‘interactivity’ functions, I suggest, not just in relation to notions of the visitor as active consumer and learner, but in terms of a project which centres around a particular vision of the relation between humans and machines. Opened in 1986, La Villette was one of a number of other major construction projects—including the Bastille opera house and the Beaubourg—which dominated Parisian cultural policy in the 1970s and 1980s. Whereas, in the UK, the imperatives of cultural policy became increasingly understood in terms of notions of consumer demand and commercial viability, the ostensible objective of the developments in Paris was to broaden public participation in culture. As Nathalie Heinich has noted, one goal of the Beaubourg project was to ‘democratise culture’ and to somehow ‘reconcile the imperatives of mass consumption with “higher” cultural production’ (Heinich 1988:199–200). Likewise, a key aim of the development of the site at La Villette was to enable a larger public to recognize the value and experience the excitement of science. Thus, the public would come to place as much value in science and technology as the French State itself. [14]

In practice, the techniques used at La Villette to encourage public interest in science obey no simple logic and are, no doubt, contradictory. On the one hand, the Cité tries to go to meet the wider publics taste not just through *vulgarisation*, but, by blurring the traditional boundaries between education and popular culture. Echoing the philosophy of many late nineteenth-century exhibitions, science is presented not just as knowledge but also as spectacle and entertainment:

Above all the Cité des Sciences et de l’Industrie is a place to learn and a place to have fun.

La Villette: a new way of seeing, listening, learning of amazement and emotion! A place for creativity and leisure, for discovery and play.

(Cité: n.d.)

On the other hand, La Villette is not entirely without the conventional marks of cultural capital. In the park outside the Cité, there are a number of ‘deconstructionist’ architectural follies designed by Bernard Tschumi (Derrida 1986), an experimental postmodern garden, a research centre for the history of science and technology, as well the new national conservatory of music—‘a complex conceived as a stimulating

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

environment and meeting place for the arts, sciences and music' (Cité n.d.).[15] Even in the Cité itself there is a multimedia library, an international conference centre and associated information services. In addition, mirroring the philosophy of the Exploratorium, is the 'Experimental gallery' which exhibits 'initiatives in art'.

the Experimental gallery exhibits artwork closely related to the fields of science, technology and industry. Though these works are neither illustrative nor educational in nature, they do represent a certain poetry, myth, humour and even criticism. The artists do not share the same views on the world as do scientists but instead provide an answer to these views.

(Cité 1988:30)

At first sight, La Villette's gestures towards art and its flirtations with post-modernism appear to mirror the Beaubourg's enthusiasm for technology. Since its opening, for example, the Beaubourg has been associated with Pierre Boulez's Institut de Recherche et de Coordination Acoustique/Musique (IRCAM), a centre which carries out research in avant-garde computer music (Born 1995). And in 1984/5 the Beaubourg supported experiments in collective computer writing in an exhibit entitled *Les Immatériaux*, which provided a vehicle for the philosopher Jean-François Lyotard to speculate about the impossibility of consensus through communication (Poster 1990:114).

Yet despite the apparent parallels between the cultural strategies of the two institutions their broader ideological resonances are quite different. In the case

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

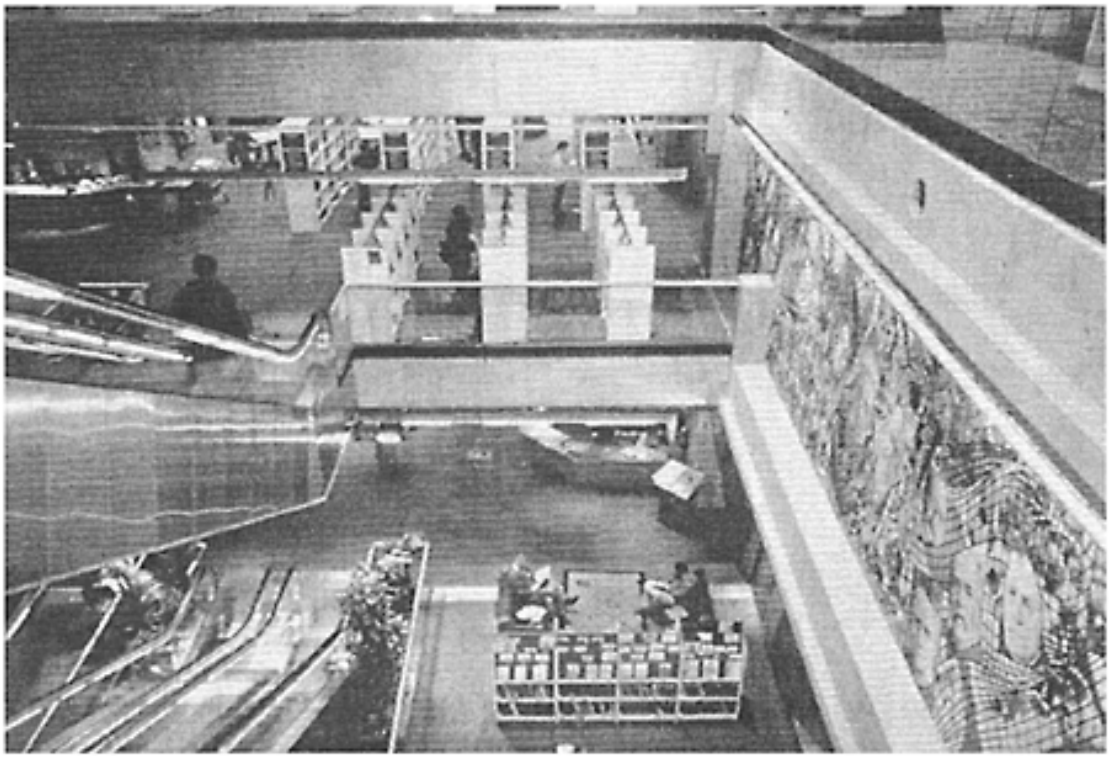

Figure 6.1 La Médiatheque: Cité des Sciences et de l'Industrie (photograph by Jean-Yves Gregoire, courtesy of the Cité des Sciences et de l'Industrie)

of the Beaubourg, the centre's close relation to technology serves to legitimize its identity as an innovative cultural institution—reinforcing the image given to it by Richard Rogers' bold architectural design. By comparison, in the case of the Cité, 'art' will always remain at the margins of an establishment which is dominated by a vision of the information age, and which tries too hard to be futuristic. Ironically, it is the Cité which appears to be the more conventional of the two institutions: its high-tech structure immediately conjuring up not so much an image of innovation and creativity, but a history of so many earlier exhibitions and philosophies in which progress has been equated with technological change.

Although the Cité's relation to the future is only too familiar, its representation of technology is none the less distinctive. Whereas the museums of the nineteenth century articulated the evolutionary metaphors of biology and political economy, the Cité's taxonomies draw on the new ahistorical sciences of communications theory, cybernetics, psychology and ecology (cf. Jordanova 1989:23). Exhibit areas are devoted to a whole series of topics concerned with the bodily and perceptual capacities of humans: sound, vision, light games, the representation of space, expression and behaviour. Moreover, although the Cité does possess the shiny rockets and cars to be found in all traditional science museums, its

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

dominant images are those associated not so much with hardware, but with language, software and the metaphors of the ‘information society’. This is a museum of information, networks, environment, multimedia, interfaces and participation (Cité 1995:23).

In practice, these ideas are manifested in a number of different ways. At the most basic level, the Cité is full of ‘interactive exhibits and audio-visual presentations’, ‘computer-based displays and games’, ‘participative, hands-on displays’ and ‘state-of-the-art museum technology’. Each of these devices has, no doubt, a specific didactic function and entertainment value. But collectively, the museum’s interactive media also have a metonymic effect. As one curator put it to me, alluding to McLuhan, ‘the medium *is* the message’. At La Villette, the future *is* interactive. Visitors to the museum purchase not a ticket but a machine readable smart card on which is written the demand ‘Découvrez!’.

However, interactive technologies do not simply function as rhetorical tropes. They also serve to organize the internal space of the museum. As Roger Silverstone has argued, the visitor’s experience of a museum may be understood as a narrative in space, the structure of which is governed, but not determined, by the spatial organization of the museum itself. This idea, derived from the work of Michel de Certeau, ‘encourages us…to begin to analyze the rhetorical and narrative strategies which are present both in an exhibition’s layout and in the routes which individuals construct through it’ (Silverstone 1988: 235). In the case of the Cité, the internal space of the museum apparently takes a quite conventional form: the visitor is guided around a three-dimensional space divided into exhibitions, shops, galleries and cafés. However, the existence of ‘interactive’ devices and technologies creates discontinuities in this space,

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

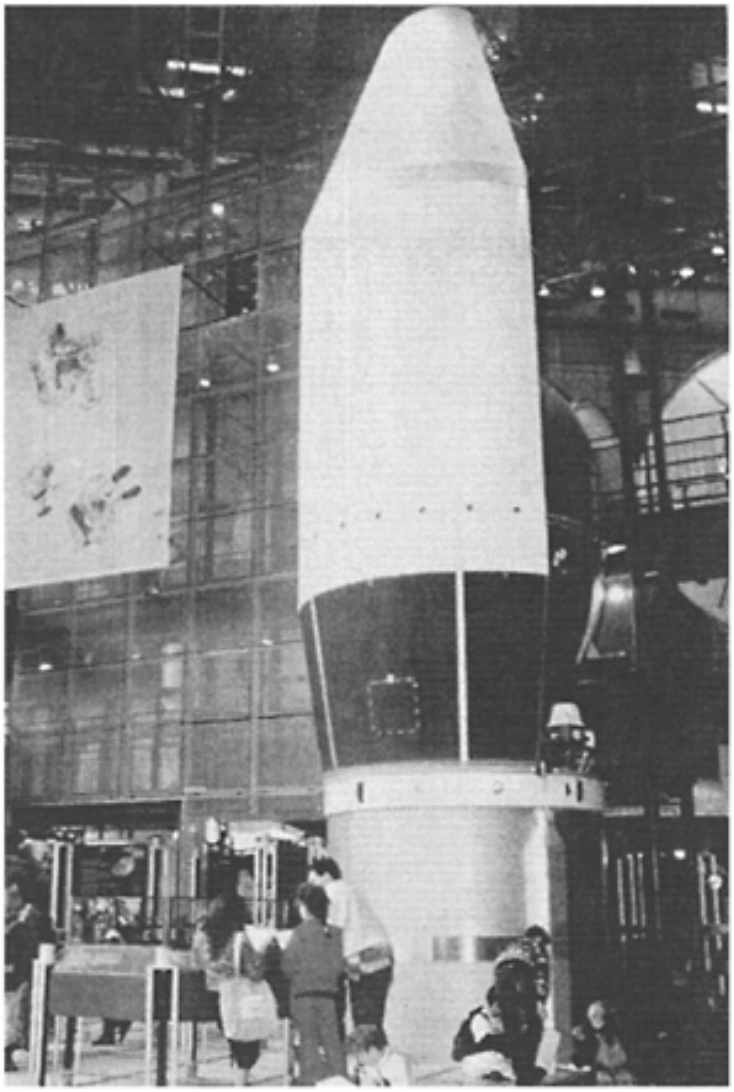

Figure 6.2 Discover! Flight simulators in the Cité (photograph by Bernard Baudin, courtesy of the Cité des Sciences et de l'Industrie)

puncturing the visitors route and establishing a further 'fourth' audio-visual dimension within which the visitor is encouraged to place him or herself, to participate and to interact. Thus, the visitor is not simply an observer of the museums machines—he or she is positioned within them. In the 'Sound' exhibit area, for example, a computer game called a 'voice-actuated note-gobbler' serves to display the tone of a persons voice. The 'Light games' area includes a section devoted to the explanation of interference which brings together a number of 'hands-on' displays. In the 'Aeronautics' area some of the most popular exhibits are flight simulators. In the 'Environment' area computer based interactive multimedia allow the visitor to explore topics such as greenery, air and trees.

Just outside of the Cité, the position of the museum visitor in the museum's exhibitionary strategy is dramatically symbolized by a huge 3-D Omnimax cinema—La Géode—in which Visual effects combine with sound effects to transport the spectators into the midst of the action surrounding them' (Cité 1988:

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

54). Reflecting on the significance of La Géode, Paul Virilio reminds us that ‘the fusion/confusion of camera, projection system and auditorium in the Imax/omnimax process, is part of a long tradition of “mobile framing” in cinema, dating from the invention of the tracking shot in 1898’ (Virilio 1990:173). Placing the Omnimax in relation to the early history of cinema is certainly appropriate. Like the cinema of the 1890s and 1900s, contemporary IMAX/ Omnimax cinema is less concerned with narrative, than with exhibition, spectacle and affect (Elsaesser 1990). However, the economic conditions of IMAX and early cinema are quite different. Whereas small-scale production companies played an important role in the development of early cinema, the relative scarcity of IMAX/Omnimax auditoria and the expense of film production have meant that the development of IMAX/Omnimax depends on corporate sponsorship (Wollen 1993). In La Géode, one popular programme is a film of the Space Shuttle produced by NASA and the Lockheed Corporation. The Space Shuttle is a particularly appropriate subject at the Cité for its design is based on the view that it *matters* that research in space depends on the involvement of humans and does not rely solely on the operation of remote-controlled instruments. In the Shuttle, humans are ‘explorers’ pushing back the final frontier of space; they have the *Right Stuff.* However, equally significantly, the physical and perceptual capacities of their own bodies are the objects of the Shuttle scientists’ experiments. Thus, at least some of the experiments performed in the Shuttle bear some comparison to those that might be found in the main body of the museum.

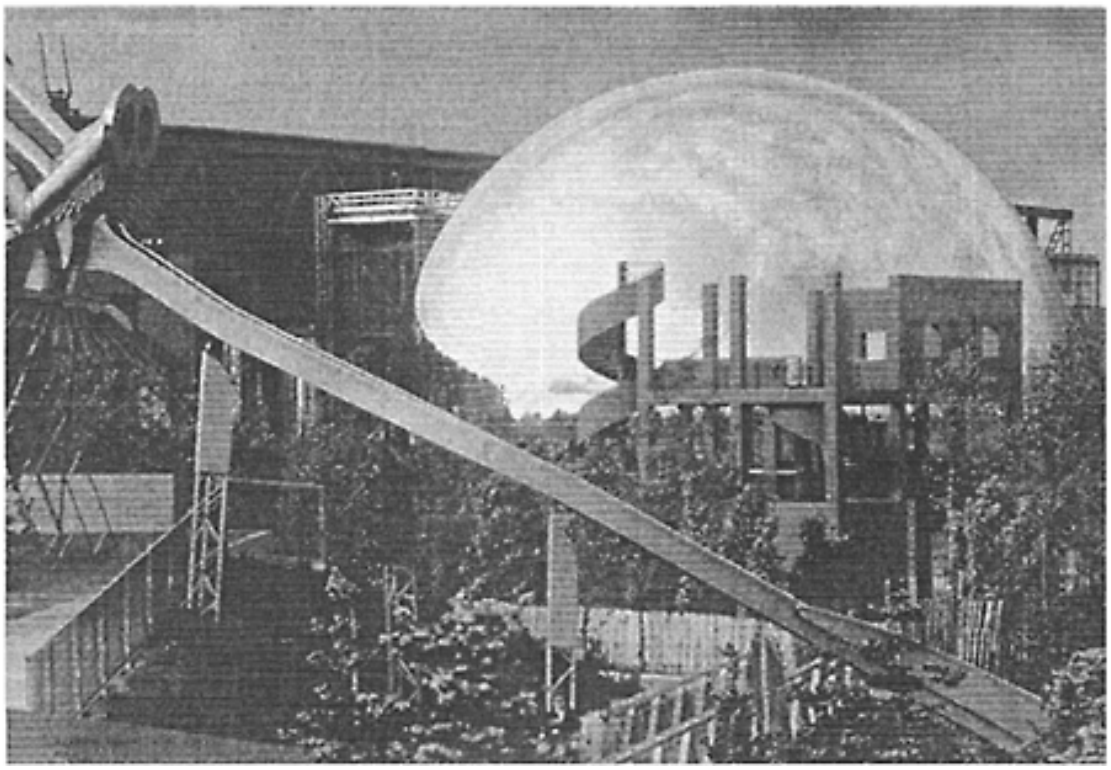

Figure 6.3 La Géode: futurism in the Cité (photograph by Arnaud Legrain, courtesy of the Cité des Sciences et de l’Industrie)

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

In the Cité the idea that science and technology reconfigure the boundaries between humans and non-humans is a pervasive one—represented not just in the ubiquity of 'interactive' techniques but in the vocabulary and taxonomy of the museum's exhibits. The Earth is understood as a 'machine' and as a 'spaceship'. Computers can 'talk'. 'Animal and vegetable kingdoms come to life in the form of automatons'. Robots and humans live in a 'cybernetic zoo'. And marriage is presented in terms of notions of 'trade' and 'system'.[16]

According to Donna Haraway, there is a tremendous political potential in the development of contemporary communication and bio-technologies. To be sure, she notes, such technologies have a crucial role in the conduct of modern states, multinational corporations, military power and labour-control systems (Haraway 1991:165), but they also render obsolete the binary oppositions between organisms and machines, revealing that 'there is no fundamental, ontological separation in our formal knowledge of machine and organism, of technical and organic' (ibid.: 178). Thus, the idea of the cyborg provides the basis for challenging the essentialism of much contemporary political thought.

Perhaps. A visit to the Cité suggests, however, that rather less productive possibilities may often be associated with the fashionable figure of the cyborg. Instead of exploring the changing role of technologies in the constitution of the social and the individual, the Cité's exhibits freeze the relation in a series of disassociated and reified 'interactions'. Thus, in a double movement, science is popularized but, at the same time, mystified. Ironically too, although it has made huge investments in media and information technologies, the Cité often fails to make any reliable connections with debates about science in the world outside; and there is little sense of the agonistic relations which invariably mark scientific and technological change.[17] In brief, at La Villette, the interaction between humans and machines can all too easily become yet another object onto which celebratory high-tech fantasies can be projected.

CONCLUSION

There are two levels of criticism that can made of the development of interactivity in the contemporary science museum. At one level, a series of questions have been asked about the use and effectiveness of interactives, not least by museum professionals and interactive designers themselves. As critics and researchers have pointed out, many interactives function as simple amusements or distractions and/or experimental games for the scientifically well-educated. However, the force of such criticisms has not

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

been any decline in interest in interactives. On the contrary, criticism has provoked professionals to improve the design of interactive devices, tailoring their design more closely to the needs, capacities and behaviour of *real* museum visitors, and integrating them more carefully with the more traditional text- and object-based exhibits. In brief, the notion of interactivity has come to be the centre of a rapidly expanding cycle of intellectual, financial and technical investments in the public presentation of science and technology.

At another level, it is important to interrogate the forms of political reasoning which have justified this remarkable level of investment in interactivity. According to Michel Callon 'technical objects…more or less explicitly define and distribute roles to humans and non-humans' (Callon 1991:137). Certainly, some of those associated with science museums and science centres have hoped that interactive devices could serve both to distribute roles to humans and to generate particular human capacities. Indeed, interactives have functioned as a kind of technical solution to the various problems that have emerged around the relations between science and the public. At the Exploratorium, for example, interactives were conceived of as ways of disseminating a sense of scientific experimentation to the wider public. In turn, the capacity to be an experimenter was taken to be equivalent to democratic empowerment. At La Villette, the idea and technology of interactivity connects together, in an ambitious project, the body of the individual visitor with a fantastic vision of a technological nation.

In contrast to France and the USA, British political and intellectual culture often appears be more ambivalent towards science and technology.[18] British critics have argued persuasively that museums should attempt to represent science and technology as complex political and cultural objects: as activities marked by contestation, uncertainty and undecidability. Indeed, far from emphasizing the importance of new forms of interactivity, such arguments point to the continuing centrality of *collections* to the public function of the contemporary museum of science.[19] According to Jim Bennett, although 'many of our former collection displays were uninspired, unimaginative and unchallenging …collections are the foundations of all great museums' and they should continue to have a central place in museum display (Bennett, Chapter 10). Although not entirely ignored, however, efforts to *rethink* the function of the museum collection have been secondary to a rather different set of concerns with questions of consumer choice and visitor behaviour. It has been the physical and perceptual capacities of the visitor, along with the development of new technologies of interactivity, which have come to be one of the dominant points of reference in contemporary debates about the design, function and future of the modern museum of science.

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

ACKNOWLEDGMENTS

Thanks to Sharon Macdonald and Roger Silverstone for their comments; to Stephen Johnston for his help and advice; and to all the staff at the Science Museum and La Villette who helped me with this research.

NOTES

1 According to one recent commentator: ‘Museums are…inherently interactive multimedia. The visitor is in control of the paths along which they navigate through the artifacts, images, sounds and texts and they are under no obligation to follow the linear structure imposed by the curator’ (Bearman 1993:183).

2 Unlike the traditional science museum, the typical science centre does not house an historical collection but is likely to rely heavily on the use of interactive exhibits. For a discussion of the difference see Durant (1992).

3 Throughout this article I refer to the National Museum of Science and Industry (NMSI) as the Science Museum.

4 For a discussion of the history of the science museum in the eighteenth century see Hooper-Greenhill (1992). According to Schaffer the public presentation of science sometimes involved an extraordinary level of public debate: natural philosophers competed for patronage and audiences and ‘critics sought to subvert the status of the lecturer’s enterprise’ (Schaffer 1993:490).

5 See for example the article by the Science Museum director Dr Neil Cossons in the *Listener* (1987). According to one member of the museum staff at the time ‘The inference in the article that the staff in the national museums are a load of dinosaurs with uncaring attitudes to the public was not well received’ (letter to the author, 30 June 1987).

6 A combination of increasing attention to marketing and the development of a public controversy about imposing museum entry charges has given the Science Museum a higher public profile (cf. Cossons

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

1991:185). However, its activities probably draw much less public comment than other museums of comparable size. As one museum curator noted: ‘what is done by the National Gallery, the V&A, the Tate etc. is always subject to both media hype *and* informed comment, from layman and specialist alike. Alas the same cannot be said of the Science Museum’ (letter to the author 18 June 1987; emphasis in original). The silence of the media and the public in relation to the politics of the Science Museum appears to be inversely related to the noise generated by visitors. According to one souvenir guide, ‘[The Science Museum] is somewhere where people feel free, and often excited; where they talk loudly (sometimes too loudly) and even laugh. It is different from most museums’ (Riemsdijk 1980:1).

7 According to Hein there were two important influences on Oppenheimer’s thinking. One was the London Science Museum Children’s Gallery (1936–94), which contained exhibits which could be operated by the child. The other was the Palais de la Découverte which, unlike traditional science museums, did not concern itself primarily with the preservation of artefacts. Created in 1937, the Palais described itself as ‘a scientific cultural centre’ in which a large number of scientific experiments were (and still are) demonstrated to visitors (Hudson 1987:103). According to its founder, Jean Perrin, one of the objectives of the Palais was to realize the potential for scientific research which he hoped might be found in the population at large. For Perrin, those young people who had not been favoured by a good education, but who had a particular aptitude for research and who had enough enthusiasm and energy to make it their vocation, should be recognized and encouraged by the National Research Service (Maury 1994:24).

8 For a overview of contemporary museum visitor studies see Bicknell and Farmelo (1993).

9 One widely cited example of such an accusation is Shortland (1987). One feature of this denigration of computer-based interactive museum exhibits is their association with interactive computer games. As Leslie Haddon observes ‘moral panics about games, including fears of addiction, the “effects” of desensitisation and of escapism have spanned a range of political campaigns, media attention and academic, mainly psychological analysis’ (Haddon 1993:124).

10 One member of the Science Museum education staff remembered the example of an interactive where a light signal was interpreted by many visitors as the cause rather than the effect of the phenomenon that the interactive was meant to demonstrate. Another suggested that many scientific principles which are

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

supposed to be revealed by interactives would only be comprehensible by A-level students and above (interviews conducted at the Science Museum, London, June 1995).

11 As Sharon Macdonald suggests, the failure of interactives to communicate scientific principles may, in part, reflect the limitations of the museum as a medium: ‘museums might not be particularly good…at getting across scientific facts and details, than furthering understanding through more general images and messages about the nature of science, its possibilities, its relevance *and* its limitations’ (Macdonald 1992:408; emphasis in original).

12 In the Science Museum there was considerable disagreement about whether this should be considered a problem or not. Proponents of interactivity noted that the Museum had a longstanding interest in interactivity from the opening of the Children’s Gallery (1936) onwards. Many others thought that integration of interactives with historical objects could be a problem as it raises questions, for example, about how to define the boundaries between objects with which the public could and could not interact. More generally, the development of interactivity in the museum not only raised questions about the function of interactive devices but also about the function of traditional objects. On these points my thanks to Sharon Macdonald, Stephen Johnston and Gillian Thomas.

13 In Lash and Urry’s (1994) terms the museum began to be engaged in a rather extensive process of reflexive modernization. In this process, the museum visitor was not necessarily conceptualized as an individual consumer. Many visitors came as part of family groups or in school parties and visitor research and exhibition design has to take this into account. In addition, the museum was aware that many of its adult visitors were male and middle class and, in this context, issues of class and gender have become a feature of exhibition design. A detailed examination of the ways in which different designs of interactive exhibits addressed specific kinds of museum visitor is beyond the scope of this chapter.

14 The post-war French State has, at least at the level of political rhetoric, tended to place great stress on the importance of science and technology for the modernization of France. By contrast, in the UK, interest in interactivity amongst scientists and museum staff developed at the height of what was perceived to be government hostility towards science in the mid-1980s.

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

15 The juxtaposition of scientific exhibitions and innovative architecture was not new in Paris (see Stamper 1989).

16 As Cornelius Castoriadis notes: ‘Ordinary mortals are ensnared together with Nobel laureates in the coils of a new mythology (“machines which think”, or “thought as a machine”)’ (Castoriadis 1984:230).

17 Compare this with Bruno Latour’s account of the need for democratic scientific institutions—a ‘parliament of things’ (Latour 1993:142–5).

18 As David Edgerton observes, the situation in Britain is more complicated than this. In the twentieth century an important strand of political thinking in Britain has supported the development of advanced sciences and technologies—for military purposes (Edgerton 1991).

19 For accounts of the difficulties of representing controversy in science museums see Young and Levidow (1984), Macdonald and Silverstone (1992) and Ross (1995). One senior member of the staff of the Science Museum suggested to me that it was difficult for the Science Museum to respond to contemporary public controversies because of the high standards of professionalism (and hence time and expense) which were demanded of gallery designers. For a more general discussion of the importance of controversy to the public understanding of science see Wynne (1992, 1993). One of the successful recent attempts to incorporate recent approaches to the history and sociology of science in museum design was undertaken at the Whipple Museum of the History of Science in Cambridge. (See Bennett *et al* 1993 and J.Bennett Chapter 10).

REFERENCES

Barry, A. (1995) ‘Reporting and visualising’, in C.Jenks (ed.) *Visual Culture*, London: Routledge.

Barry, A., T.Osborne and N.Rose (eds) (1996) *Foucault and Political Reason: Liberalism, Neo-liberalism and Rationalities of Government*, London: UCL Press/ Chicago: Chicago University Press.

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

Bearman, D. (1993) ‘Interactivity in American museums’, *Museum Management and Curatorship*, 12, pp. 183–93.

Bennett, J., R.Brain, K.Bycroft, S.Schaffer, H.Sibum and R.Staley (1993) *Empires of Physics*, Cambridge: Whipple Museum of the History of Science.

Bennett, T. (1995) *The Birth of the Museum: History, Theory, Politics*, London: Routledge.

Bicknell, S. and G.Farmelo (eds) (1993) *Museum Visitor Studies in the 1990s*, London: Science Museum.

Born, G. (1995) *Rationalizing Culture: IRCAM, Boulez and the Institutionalization of the Musical Avant-Garde*, Berkeley, CA: University of California Press.

Callon, M. (1991) ‘Techno-economic networks and irreversibility’, in J.Law (ed.) *A Sociology of Monsters: Essays on Power, Technology and Domination (Sociological Review Monograph)*, pp. 132–61, London: Routledge.

Castoriadis C. (1984) *Crossroads in the Labyrinth*, Cambridge, MA: The MIT Press.

Cité des Sciences et de l’Industrie, direction de la Communication et du développement (1988) *Cité des Sciences et de l’Industrie*, Paris.

(n.d.) visitor information.

(1995) E*xplora: Guide to the Permanent Exhibitions (*6th edition), Paris: Direction de la Communication et de la Promotion, la Villette.

Cossons, N. (1987) ‘Adapt or die: dangers of a dinosaur mentality’, *Listener* (16 April), pp. 18–20.

(1991) ‘Scholarship or self-indulgence’ RS*A Journal (F*ebruary) 139 (5415), pp. 184–91.

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

Cruikshank, B. (1996) ‘Revolutions within: self-government and self-esteem’, in A.Barry, T.Osborne and N.Rose (eds) *Foucault and Political Reason: Liberalism, Neo-liberalism and Rationalities of Government*, pp. 231–52, London: UCL Press/ Chicago: Chicago University Press.

Délacôte, G. (1992) http://www.exploratorium.edu/general/directors-vision.html.

Derrida, J. (1986) ‘Point de folie—maintainant l’architecture’, *AA Files*, 12, pp. 65–75.

Durant, J. (1992) ‘Introduction’, in J.Durant (ed.) *Museums and the Public Understanding of Science*, London: Science Museum.

Edgerton, D. (1991) ‘Liberal militarism and the British state’, *New Left Review*, 185, Jan-Feb.

Elsaesser, T. (ed.) (1990) *Early Cinema: Space, Frame, Narrative*, London: British Film Institute.

Foucault, M. (1979) *Discipline and Punish*, Harmondsworth: Penguin.

Galison, P. (1994) ‘The ontology of the enemy: Norbert Wiener and the cybernetic vision’, *Critical Inquiry*, 21 (Autumn), pp. 228–66.

Gregory, R. (1970) *The Intelligent Eye*, London: Weidenfeld & Nicholson.

(1989) ‘Turning minds to science by hands-on exploration: the nature and potential of the hands-on medium’, in Nuffield Interactive Science and Technology Project *Sharing Science: Issues in the Development of Interactive Science and Technology Centres*, London: Nuffield Foundation/Committee on the Public Understanding of Science.

Haddon, L. (1993) ‘Interactive games’, in P.Hayward and T.Wollen (eds) *Future Visions: New Technologies of the Screen*, pp. 123–47, London: British Film Institute.

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

Haraway, D. (1991) *Simians, Cyborgs and Women: The Reinvention of Nature*, London: Free Association Books.

Hein, H. (1990) *The Exploratorium: The Museum as Laboratory*, Washington, DC: Smithsonian Institution Press.

Heinich, N. (1988) ‘The Pompidou Centre and its public: the limits of a utopian site’, in R.Lumley (ed.) *The Museum Time-machine: Putting Cultures on Display*, pp. 199–212, London: Routledge.

Hooper-Greenhill, E. (1992) *Museums and the Shaping of Knowledge*, London: Routledge.

Hudson, K. (1987) *Museums of Influence*, Cambridge: Cambridge University Press.

Jordanova, L. (1989) ‘Objects of knowledge: an historical perspective on museums’, in P.Vergo (ed.) *The New Museology*, London: Reaktion Books.

Kirby, S. (1988) ‘Policy and politics: charges, sponsorship and bias’, in R.Lumley (ed.) *The Museum Time Machine*, London: Routledge/Comedia.

Lash, S. and J.Urry (1994) *Economies of Signs and Space*, London: Sage.

Latour, B. (1986) ‘The powers of association’, in J.Law (ed.) *Power, Action and Belief*, London: Routledge.

(1993) W*e Have Never Been Modern*, *H*emel Hempstead: Harvester Wheatsheaf.

Law, J. (1986) ‘On power and its tactics: a view from the sociology of science’, *Sociological Review* 34 (1), pp. 1–35.

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

Levidow, L. and R.Young (1984) ‘Exhibiting nuclear power: the Science Museum cover-up’, *Radical Science*, 14, pp. 53–79.

Macdonald, S. (1992) ‘Cultural imagining among museum visitors’, *Museum Management and Curatorship*, 11, pp. 401–9.

(1993) ‘Un nouveau “corps des visiteurs”: musées et changements culturels’, *Publics et Musées*, 3, pp. 13–27.

Macdonald, S. and R.Silverstone (1990) ‘Rewriting the museums’ fictions: taxonomies, stories and readers’, *Cultural Studies*, *4* (2), pp. 176–91.

(1992) ‘Science on display: the representation of scientific controversy in museum exhibitions’, *Public Understanding of Science*, 1, pp. 69–87.

Maury, J-P. (1994) *Le Palais de la Découverte*, Paris: Gallimard.

Nash, C. (1992) ‘Interactive media in museums: looking backwards, forwards and sideways’, *Museum Management and Curatorship*, 11, pp. 171–84.

Poster, M. (1990) *The Mode of Information: Poststructuralism and Social Context*, Cambridge: Polity.

(1995) T*he Second Media Age*, *C*ambridge: Polity.

Reichardt, J. (ed.) (1971) *Cybernetics, Art and Ideas*, London: Studio Vista.

Riemsdijk, J. van (1980) *The Science Museum*, London: The Science Museum.

Rose, N. (1993) ‘Government, authority and expertise in advanced liberalism’, *Economy and Society*, 22 (3), 283–99.

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

(1996) ‘Governing “advanced” liberal democracies’, in A.Barry, T.Osborne and N.Rose (eds) *Foucault and Political Reason: Liberalism, Neo-liberalism and Rationalities of Government*, London: UCL Press /Chicago: Chicago University Press.

Ross, M. (1995) ‘Museums and controversy: the case of passive smoking’, *Science as Culture*, 5 (1), pp. 147–51.

Schaffer, S. (1992) ‘Self evidence’, *Critical Inquiry* 18, pp. 327–62.

(1993) ‘The consuming flame: electrical showmen and Tory mystics in the world of goods’, in J.Brewer and R.Porter (eds) *Consumption and the World of Goods*, London: Routledge.

Science Museum: the National Museum of Science and Industry (1986), ‘Management Plan’, London.

Shortland, M. (1987) ‘No business like show business’, *Nature*, 328, p. 213.

Silverstone, R. (1988) ‘Museums and the media: a theoretical and methodological exploration’, *The International Journal of Museum Management and Curatorship*, 7, pp. 231–41.

(1992) The medium is the museum: on objects and logics in times and spaces’, in J.Durant (ed.) *Museums and the Public Understanding of Science*, London: Science Museum.

Stamper, J.W. (1989) The galerie of machines of the 1889 Paris World’s Fair’, *Technology and Culture*, 30 (2), pp. 330–53.

Stevenson, J. (n.d.) ‘The philosophy behind launch pad’, Science Museum, *Journal of Education and Museums*, 8, pp. 10–18.

(1994) ‘Getting to Grips’, *Museums Journal*, *M*ay, pp. 30–2.

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.

Copyright © 1998. Taylor & Francis Group. All rights reserved.

Strathern, M. (1992) *Reproducing the Future: Anthropology, Kinship and the New Reproductive Technologies*, Manchester: Manchester University Press.

Thomas, G. (n.d.) 'The National Museum of Science and Industry: facing the future', mimeo.

Virilio, P. (1990) 'Cateract surgery: cinema in the year 2000', in A.Kuhn (ed.) *Alien Zone: Cultural Theory and Contemporary Science Fiction Cinema*, London: Verso.

Wollen, T. (1993) 'The bigger the better: from cinemascope to IMAX', in P.Hayward and T.Wollen (eds) *Future Visions: New Technologies of the Screen*, pp. 10–30, London: British Film Institute.

Wynne, B. (1992) 'Misunderstood misunderstandings: social identities and the public uptake of science', *Public Understanding of Science*, 1, pp. 281–304.

(1993) 'Public uptake of science: a case for institutional reflexivity', *Public Understanding of Science*, 2, pp. 321–37.

The Politics of Display : Museums, Science, Culture, edited by Sharon Macdonald, Taylor & Francis Group, 1998. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/anu/detail.action?docID=618865.
Created from anu on 2025-07-28 03:46:45.